# CSE 350

## Digital Electronics and Pulse Techniques

## Assignment-03

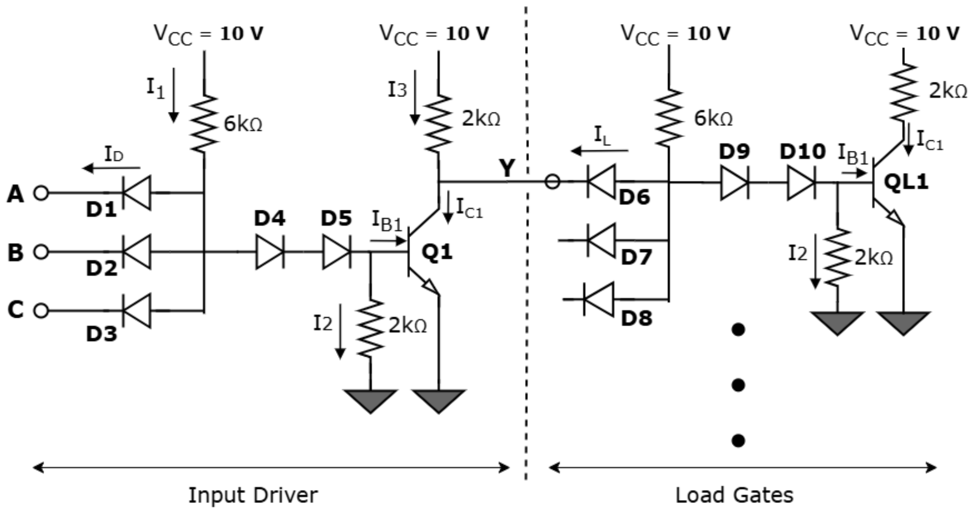
**Total – 20 marks**

### Question 1 — 05

Implement the logic function, $Y = \overline{A+B}\,.\,\overline{C}$ using **any combination of the logic families** you know. Then, mention the **operating mode** of your diodes/transistors when the **output is high**. [5]

[*You may simplify the function if possible. Mentioning the values of circuit components is not necessary.*]

### Question 2 — 05




For the above circuit, $B_F = 30$, $V_{CE}$ (sat) = 0.1 V, $V_{BE}$ (sat) = 0.8 V [1+2+2]

(a) Find $\mathbf{I_1, I_2, I_{B1}, I_{C1}}$ and $\mathbf{I_D}$ if any input is logical low (0.1 V) for the driver circuit when **load gates are not connected**. Then also find out the power dissipation (of the driver) for this case.

(b) Find $\mathbf{I_1, I_2, I_{B1}, I_{C1}}$ and $\mathbf{I_D}$ if all input is logical high (10 V) for the driver circuit when **load gates are not connected**. Then also find out the power dissipation (of the driver) for this case. Prove that Q1 transistor is in Saturation mode.

(c) The driver circuit is now connected with Load gates. Calculate maximum fanout of the overall DTL setup.

$\boxed{\text{Set A}}$ (a) let's say 'A' is low $\longrightarrow V_A = 0.1\ V \Longrightarrow$ '$D_1$' is conducting

so, $V_P = 0.1 + 0.7 = 0.8V$

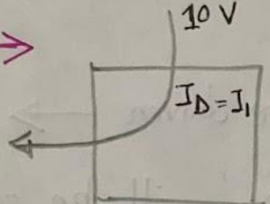
$\therefore I_1 = \dfrac{10-0.8}{6k} = 1.533\ mA = I_D$

due to $V_P = 0.8V$;
$D_4, D_5$ turned off
$Q_1 \rightarrow$ cut off

$\therefore I_2 = 0\ mA$

$I_{B_1} = 0\ mA \quad ; \quad I_{C_1} = 0\ mA$

Power dissipation $\longrightarrow$




$P = (10-0.1) \times 1.533$
$= 15.18\ mW$

(b) $V_A = V_B = V_C = 10\ V \Rightarrow D_1, D_2, D_3$ all turned off $\Rightarrow \boxed{I_D = 0}$

so, $V_P = 0.7 + 0.7 + 0.8 = 2.2\ V$

$\therefore I_1 = \dfrac{10-2.2}{6k} = 1.3\ mA$

$I_2 = \dfrac{0.8-0}{2k} = 0.4\ mA$

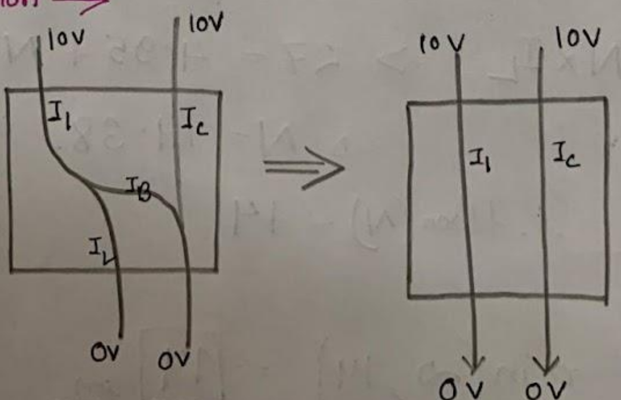
$\therefore I_{B_1} = I_1 - I_2 = (1.3 - 0.4) = 0.9\ mA$

$D_4, D_5 \rightarrow$ conducting
$Q_1 \longrightarrow$ saturated

$I_{C_1} = \dfrac{10-0.1}{2k} = 4.95\ mA$

$\therefore \beta_{forced} = \dfrac{I_{C_1}}{I_{B_1}} = \dfrac{4.95}{0.9} = 5.5 < \beta_F$ (thus, '$Q_1$' is in saturation mode)

power dissipation $\rightarrow$




$P = P_1 + P_2$
$= (10-0) \times 1.3 +$
$(10-0) \times 4.95$
$= 62.5\ mW$

(c) case → any 'input' of the driven low → will lead to $V_y$ being 'High' → thus '$D_6$' will not be conducting.

so, $I_L = 0\ mA$ current flowing into the driven circuit from loads.

so, we can connect as many loads as possible in such case.

Fanout $= \infty$

case → all 'inputs' high of the driven → will lead to $V_y$ being 'low' → thus '$D_6$' will be conducting.

For each load, we consider the (011) case, so as a result, Maximum current will flow in '$D_6$' path and to our driven circuit.

$V_y = 0.1$, then, $V_{P\,load} = 0.1 + 0.7 = 0.8$

$\therefore I_L = \dfrac{10 - 0.8}{6k} = 1.533\ mA$

let's say, there are 'N' numbers of loads connected

$I_3 = \dfrac{10 - 0.1}{2k} = 4.95\ mA$

using constraint rule, $I_{cmax} = \beta_F \times I_{B_1} = 30 \times 0.9 = 27\ mA$

$\therefore I_{cmax} = I_3 + N \times I_L \Rightarrow 27 = 4.95 + N \times 1.533$

$\Rightarrow N = 14.38$

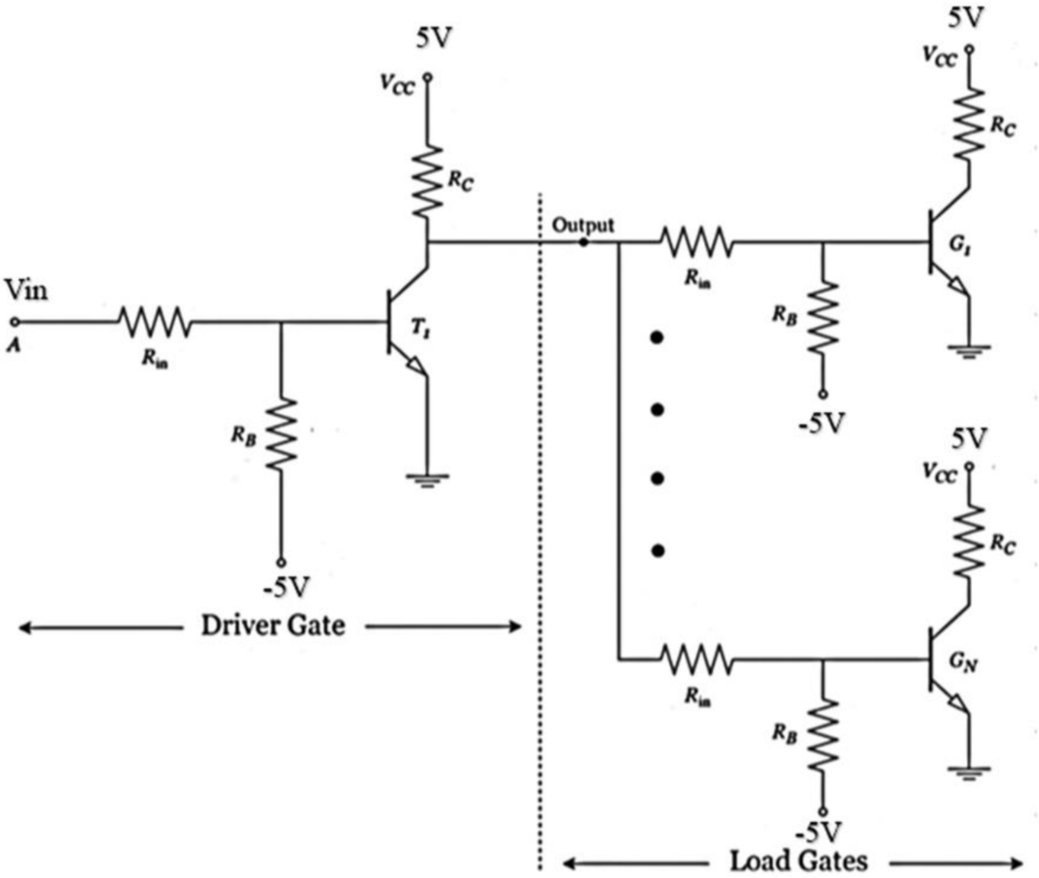
$\therefore floor(N) = 14$

$\therefore$ fanout $= 14$

$\therefore max^m$ fanout $= min(\infty, 14) = \boxed{14}$ ans

## Question 3 **05**

For the given circuit assume $V_{OH}$ = 4.5V and $V_{OL}$ = 0.2V. Also assume common emitter current gain, $\beta_F$ = 30. $V_{BE}$(sat) = 0.8 V, $V_{CE}$(sat) = 0.2 V and cut in voltage for transistor $V\gamma$ = 0.5 V.




Driver and Loads are identical where **$R_{in}$ = 5k**, **$R_B$ = 100k**, and **$R_C$=2k.**

| | | |
|---|---|---|
| (a) | If **Vin = High**, find the power dissipation in the **Driver** circuit assuming **48 loads** are connected. | 2.5 |
| (b) | Find the value of **Noise margin** of the circuit. | 2.5 |

## Set-A (Solⁿ)

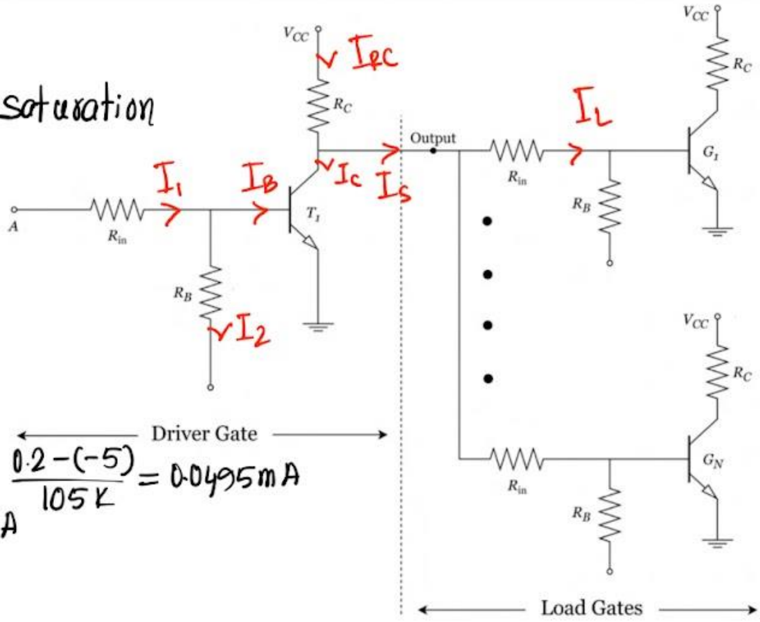
(a) $V_{in} = high = 5V \rightarrow T_1 \rightarrow$ saturation

$V_B = 0.8V,\ V_C = 0.2V$

$I_1 = \dfrac{5-0.8}{5K} = 0.84mA$

$I_2 = \dfrac{0.8-(-5)}{100K} = 0.058mA$

$I_B = I_1 - I_2 = 0.782mA$



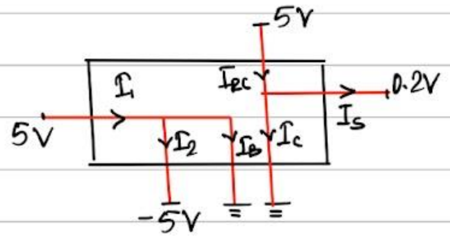

Load Transistors $\rightarrow$ cutoff

individual load current, $I_L = \dfrac{0.2-(-5)}{105K} = 0.0495mA$

total $I_S = 48 I_L = 2.376mA$

$I_{RC} = \dfrac{5-0.2}{2K} = 2.4mA$

$\therefore I_C = I_{RC} - I_S = 0.024mA$




$P = (5+5)I_2 + (5-0)I_B + (5-0.2)I_S + (5-0)I_C$

$\boxed{P = 16.01\ mW}$

(b) $V_{IL}$ calculation

$T_1$ is on the verge of turn on $\rightarrow V_B = V_\gamma = 0.5V$

$I_2 = \dfrac{0.5+5}{100K} = 0.055mA$

$I_1 = I_2 = 0.055 = \dfrac{V_i - 0.5}{5K}$

$\Rightarrow V_i = 0.775V$

$\therefore V_{IL} = 0.775V,\ V_{OL} = 0.2V \rightarrow \boxed{V_{NL} = V_{IL} - V_{OL} = 0.575V}$

$V_{IH}$ Calculation

$T_1 \rightarrow$ sat, on the verge of saturation and f. Active.

$\therefore \beta_{forced} \approx \beta_F = 30,\ V_B = 0.8V,\ V_C = 0.2V$

$I_C = \dfrac{5-0.2}{2K} = 2.4mA,\ \beta = \dfrac{I_C}{I_B} \Rightarrow I_B = \dfrac{I_C}{30} = 0.08\ mA$

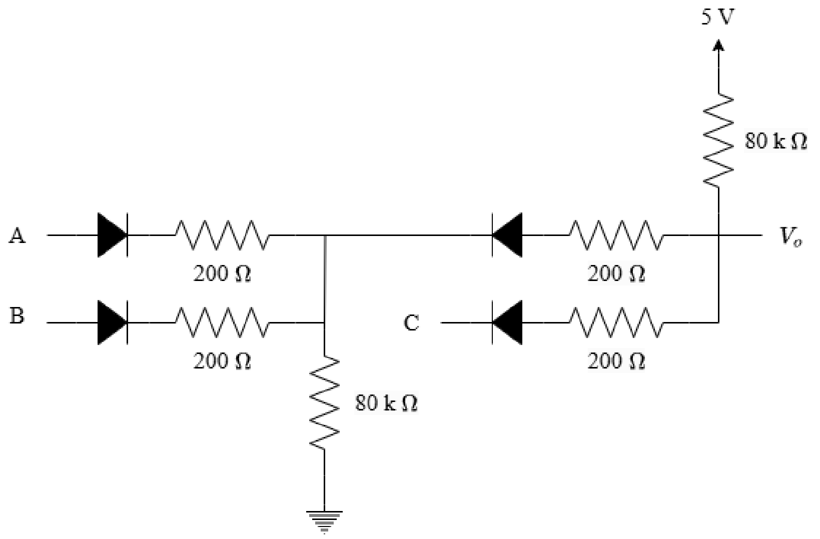
$I_2 = \dfrac{0.8+5}{100K} = 0.058mA,\ I_1 = I_2 + I_B = 0.138mA$

$I_1 = 0.138 = \dfrac{V_i - 0.8}{5K} \Rightarrow V_i = 1.5V$

$\therefore V_{IH} = 1.5V,\ V_{OH} = 4.5V \quad \therefore \boxed{V_{NH} = 4.5 - 1.5 = 3V}$

$\therefore$ Noise Margin $\rightarrow NM = \min(V_{NL}, V_{NH}) = 0.575V$

## Question 4 05




For the circuit shown, input voltage,

$V_{in}(high) = 5\ V, V_{in}(low) = 0\ V$

| | | | |
|---|---|---|---|
| (c) | CO2 | What logic function does the above circuit implement? | [2.5] |
| (b) | CO2 | Find the output voltage & power dissipation of the entire circuit when A=low, B=high, C=low | [2.5] |

**Solution:**

a) $I_6 = I_3 + I_{B3} = I_3 + I_4 - I_5 = I_3 + I_{B_2} + I_2 - I_5$

$= I_3 + I_1 + I_{E_A} + I_{E_B} + I_2 - I_5$

$\rightarrow 4.74 = \frac{6-0.2}{4.5} + \frac{(6-2.3)}{4.5} + 0.2 \times \frac{6-2.3}{4.5} \times 2 + \frac{6-1}{R} - \frac{0.8}{4}$

$R = 2\ k\Omega$

b) $(A + B)C$

c) KCL at output, $\frac{5-V_o}{80} = \frac{V_o-0.7}{0.2} \rightarrow V_o = 0.71\ V$

$P = (5-0) \times \frac{5-0.7-0}{0.2+80} + (5-0) \times \frac{5-0.71}{80} = 0.536\ mW$